

NS Aquaculture Has No Early Warning System

Farmers are flying blind.

\$430M

NS aquaculture industry / yr

Days

MHW can wipe out a farm in days

Cat IV

Active MHW across 4 NS sites now

THE SOLUTION

BlueFish

Real-time marine heatwave monitoring for NS aquaculture farmers



BlueFish



Live SST Tracking

Satellite sea surface temperature updated
across 4 NS monitoring sites



7-Day Forecast

Predictive outlook so farmers can plan
ahead before conditions worsen



Species Risk Panel

Salmon · Mussel · Oyster thermal stress and
mortality thresholds

From Satellite to Dashboard in 3 Steps

01

NOAA SST Data

Satellite sea surface temperature pulled via ERDDAP API. Fallback to CSV for offline reliability.

02

Hobday MHW Algorithm

Scientific detection algorithm compares SST against 90th percentile climatological threshold to classify heatwave category.

03

BlueFish Dashboard

FastAPI backend + Next.js frontend. Live alerts, site comparison, species risk, 7-day forecast — all in one view.

Runs offline · No API keys required · No cloud dependency

See It Live

[Live Demo](#)

1. Click each site card → SST and MHW status updates
2. SST chart shows 3 lines: actual temperature, threshold, and climatology
3. 7-day forecast — all 7 days red = heatwave continuing
4. Species risk panel — Atlantic Salmon showing 100% disease risk

Why It Matters

\$430M

at risk every year

Cat III–IV

= mass mortality event

7 days

advance forecast window

Early warning enables farmers to:

Harvest early · Move cages · Reduce feeding · Minimize losses